WORKED SOLUTION

STEP I 1992, Question 14

Probability and Statistics • Interpreting Statistical Claims

STEP I 1992, Question 14

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/92-s1-q14

This question has a full guided treatment in *Beyond STEP*, Section 13.1, *Reading the Headlines*: https://beyondstep.keithmcnulty.org/ch13/headlines.html

The key idea

Every headline takes one number and leaps to a conclusion. For each one, ask two questions. First, is it comparing like with like: the same thing, measured in the same way, scaled by how much exposure there is to the risk? Second, could the difference just be chance? For the second, a natural yardstick is the Poisson model for counts of rare, independent events, whose standard deviation is the square root of its mean. Knowing when that model fits, and when it doesn't, is the heart of (ii) to (iv).

Solution

Throughout, "on average" means the 1974–1989 annual averages: 1080 pedestrian deaths and 180 air deaths a year.

Part (i)

The ratio $1080/180 = 6$ is where the "six times" comes from, but it compares raw totals, not risks. A risk is deaths per unit of exposure: per journey, per hour, or per kilometre travelled. Almost everyone in Poldavia walks near roads every day, while comparatively few people fly, and then only occasionally. To compare safety we would need something like

$$\frac{\text{pedestrian deaths}}{\text{total hours (or km) walked}} \quad \text{against} \quad \frac{\text{air deaths}}{\text{total hours (or km) flown}},$$

and the headline gives us neither denominator. The answer could even go the other way, depending on the measure: per kilometre, flying looks very safe, but per hour or per journey it may look less so.

There are other problems too. "Walking" here really means being hit by a vehicle while on foot, and the figure says nothing about pedestrians in quiet places. Air deaths include crew as well as passengers. And averages over a period cannot "prove" anything about safety in 1991; at best they are evidence about 1974–1989.

Part (ii)

At first sight 125 is well below the average of 180. Is the drop more than chance?

If deaths happened independently, one at a time, a Poisson model with mean 180 would be reasonable. Its standard deviation is $\sqrt{180} \approx 13.4$, so 125 would be about 4.1 standard deviations below the mean, and very unlikely to happen by chance ($P(X \leq 125) \approx 10^{-5}$).

But air deaths do not happen one at a time. They come in clusters: a single crash can kill tens or hundreds of people. The natural model is that the number of *fatal crashes* is Poisson, and each crash kills a random number of people. This hugely increases the variability. For example, suppose there are on average 6 fatal crashes a year, each killing 30 people. Then the annual death toll is $30N$ with $N \sim \text{Po}(6)$, which has standard deviation $30\sqrt{6} \approx 73$, and

$$P(\text{deaths} \leq 125) = P(N \leq 4) \approx 0.29.$$

On those (illustrative) figures, a year with 125 deaths is entirely ordinary: it needs only one or two fewer crashes than usual. Letting the size of each crash vary as well makes the spread larger still.

So a single year's figure tells us very little. We would also want to know how much flying took place in 1990 (if there was less air traffic, fewer deaths would be expected anyway) and the figures for several years, to see whether there is a genuine trend. The headline's claim is not supported.

Part (iii)

Two separate things are wrong here.

One day cannot show an increase. "Increasing" is a claim about a trend over time, and a single day's figure gives no comparison at all. We would need daily (or annual) figures over a long period.

Seven is not even unusual. The only road figure we have is for pedestrians: 1080 a year, or

$$\lambda = \frac{1080}{365} \approx 2.96 \text{ per day.}$$

Pedestrian deaths are fairly well modelled as Poisson: they are rare, mostly one per accident, and roughly independent. Let $X \sim \text{Po}(2.96)$ be the number of pedestrians killed on a given day. Then

$$P(X \geq 7) = 1 - \sum_{k=0}^{6} \mathrm{e}^{-\lambda} \frac{\lambda^k}{k!} \approx 0.031,$$

so we would expect about $365 \times 0.031 \approx 11$ days a year on which at least 7 pedestrians die. And the headline says "people killed", which includes drivers, passengers and cyclists as well. The total daily road death rate must be higher than 2.96, which makes 7 even less remarkable.

Ask yourself Isn't a day with more than double the average number of deaths alarming?

Not for small counts. When the mean is about 3, the Poisson standard deviation is $\sqrt{3} \approx 1.7$, which is large compared with the mean, so big proportional swings from day to day are normal. Picking out the worst day is also a selection effect: there are 365 days in a year, and a newspaper can always find a bad one. The same day of the week also matters, since traffic and the number of people out on foot vary through the week and the seasons. A Poisson model with a constant rate is only a rough first approximation.

Part (iv)

This is the headline with a real case to answer. First the statistics, then the interpretation.

Is 1350 *more than chance?* Here the Poisson model is a good starting point, for the reasons given in (iii). With mean 1080, the standard deviation is $\sqrt{1080} \approx 32.9$, and

$$\frac{1350 - 1080}{32.9} \approx 8.2.$$

A value 8 standard deviations above the mean has probability of order 10^{-15} under the model. Even if the true year-to-year spread were two or three times the Poisson value (because of bad winters, say), 1350 would still be very unusual compared with a steady rate of 1080. So something has very probably changed. The question is what.

Compared with what? The headline compares 1990 with a 16-year average, not with 1989. If pedestrian deaths had been rising steadily over the period, the average of 1080 would describe the middle of it (around 1981–82), and 1990 should be well above it. For instance, a steady rise of 30 deaths a year, from about 855 in 1974, has average 1080 and gives about 1335 in 1990. In that case 1990 would be no "rise" at all, just the continuation of a long trend. That trend might deserve a headline, but a different one. We need the year-by-year figures to tell the two stories apart.

Has the exposure changed? More people, more vehicles, or more walking would all increase deaths without making walking any more dangerous per journey. The fair comparison is a rate, such as deaths per 100 000 population or per kilometre walked.

Has the measurement changed? A change in how deaths are recorded or classified would cause an apparent jump. Examples are a change in how long after an accident a death still counts as a road death, or in who counts as a pedestrian, or better reporting. Any such change would need to be ruled out.

Is the word right? "Casualties" normally means people killed *or injured*. The figure quoted is for deaths only, so the headline overstates what the number shows. Injuries are far more numerous, and they might show quite a different pattern.

In summary: the rise to 1350 is far too large to be explained by Poisson variation about a constant rate of 1080. But before calling it "alarming" we need the recent annual figures (to separate a sudden rise from a long-term trend) and the population and traffic figures (to turn counts into rates). We also need to check that the definitions have not changed.

Checks

The probabilities were checked by summing the Poisson series exactly and by simulation with 2×10^5 trials. For (iii), $P(X \geq 7) = 0.0315$ when $\lambda = 1080/365$ (simulation: 0.0317). For (ii), with 6 crashes a year of 30 deaths each, $P(N \leq 4) = 0.285$. Letting the crash size vary uniformly from 1 to 59 gave standard deviation 84 and $P(\text{deaths} \leq 125) \approx 0.28$ by simulation. A quick rule of thumb for a Poisson count is that values within about two square roots of the mean are unremarkable. For one day, $3 + 2 \times 1.7 \approx 6.4$, so 7 is only just outside that range, and with 365 days to choose from such days are bound to occur. For a year, $1080 + 2 \times 33 = 1146$, which is nowhere near 1350.

Common slips

- Comparing totals instead of rates in (i): a risk always needs a denominator.
- Using the Poisson standard deviation $\sqrt{180}$ in (ii). Deaths in air crashes are clustered, so the spread is far larger.
- Stopping at "8 standard deviations, so it's a real rise" in (iv), without asking whether the comparison with a 16-year average, the population or the definitions explain it.

Beyond STEP

Part (ii) is about *overdispersion*. A total made up of a Poisson number of clusters, each of random size, is a *compound Poisson* variable, and its variance is $\lambda\,\mathrm{E}[S^2]$, not λ. Part (iv) leads into hypothesis testing, and into the much harder question of what a statistically significant change actually means. Both appear early in a university probability and statistics course. Blitzstein and Hwang, *Introduction to Probability*, is a good place to read about Poisson models and their limits.

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